Efficient Posterior Sampling for $\mathbb{Z}_2$ Synchronization

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From statistical inference to posterior sampling

$$\boldsymbol{Y} = \frac{\lambda}{\sqrt{n}}\boldsymbol{\theta}\boldsymbol{\theta}^\top + \boldsymbol{Z}, \qquad \boldsymbol{\theta} \sim \mathrm{Unif}\{\pm 1\}^n.$$

- Noise: $\boldsymbol{Z} = \boldsymbol{Z}^\top$, independent of $\boldsymbol{\theta}$, with independent Gaussian entries above the diagonal, $Z_{ij} \sim \mathcal{N}(0,1)$ and $Z_{ii} \sim \mathcal{N}(0,2)$.

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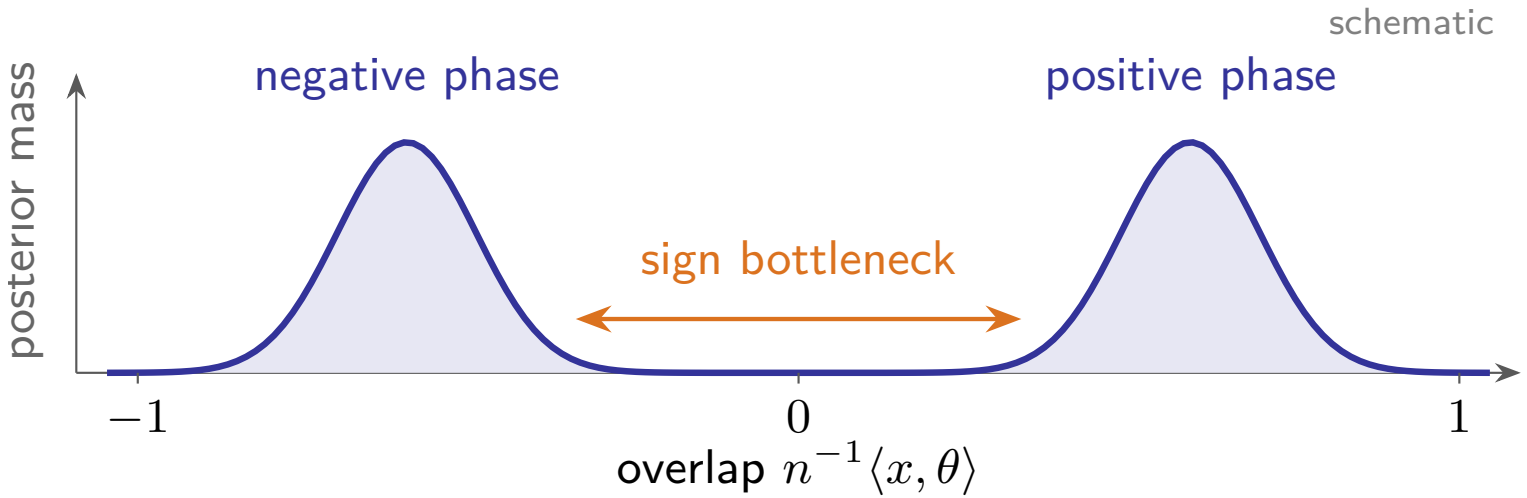
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Our question

Given $\boldsymbol{Y}$ and known $\lambda > 1$, can we efficiently draw a sample from the full posterior of $\boldsymbol{\theta}$?

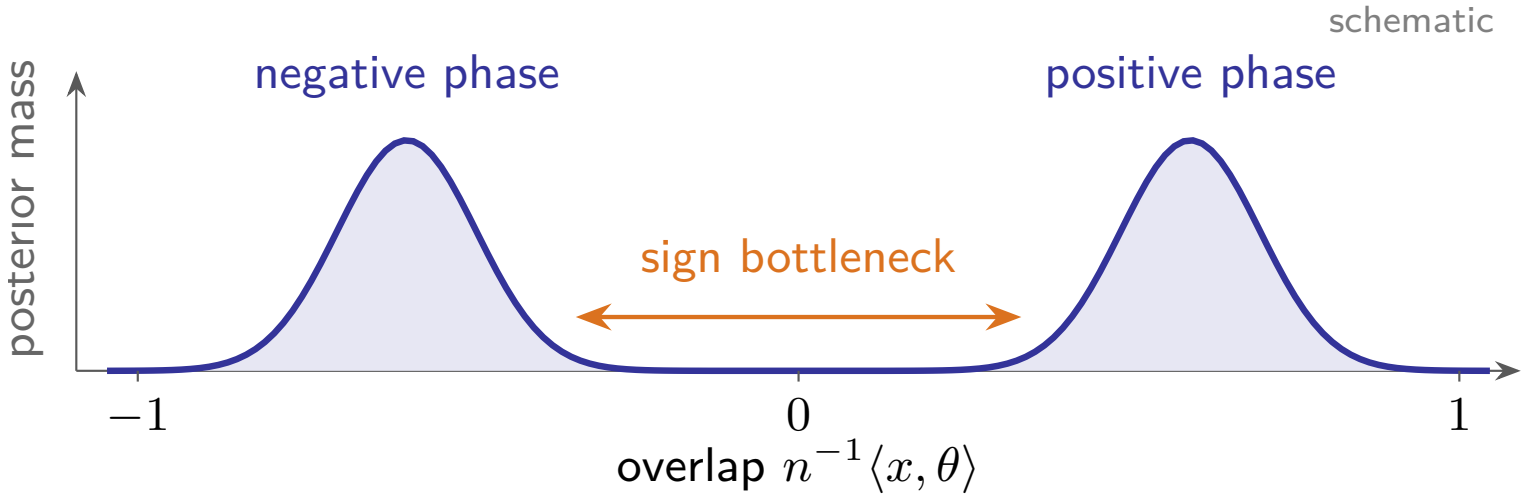
The posterior has two symmetry-related phases

$$\mu_{\boldsymbol{J}}(x) \propto e^{H_{\boldsymbol{J}}(x)}, \quad H_{\boldsymbol{J}}(x) = \tfrac{1}{2} x^\top \boldsymbol{J} x, \quad \boldsymbol{J} = \frac{\lambda}{\sqrt{n}} \boldsymbol{Y}.$$



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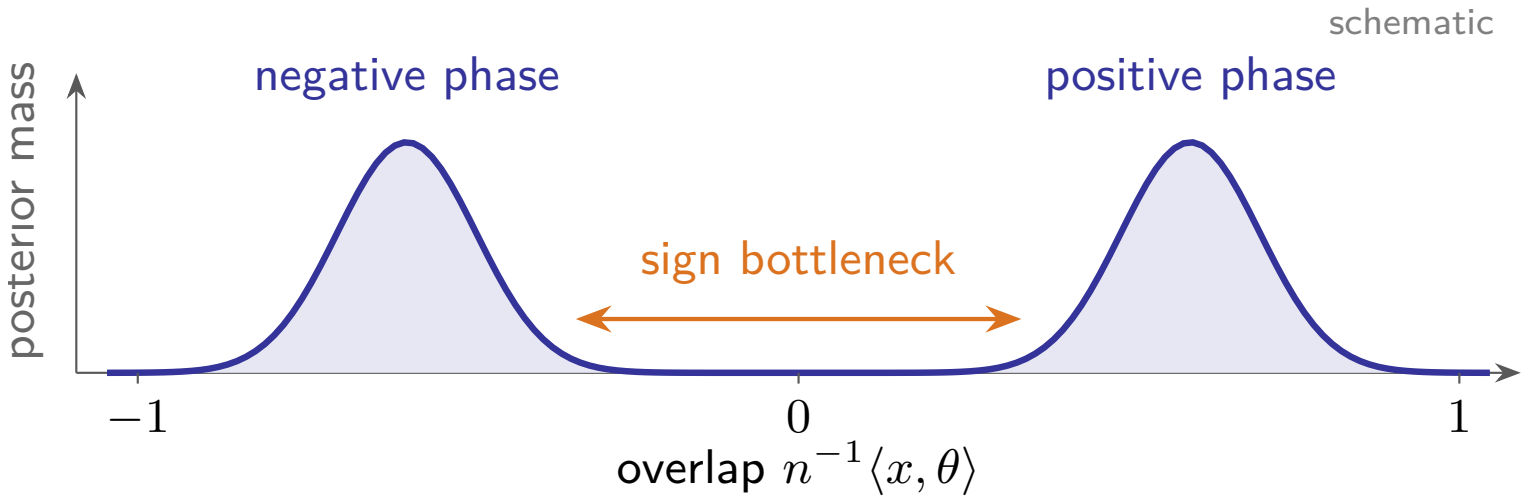
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- $\mu_{\boldsymbol{J}}(x) = \mu_{\boldsymbol{J}}(-x)$, so $\mathbb{E}[\boldsymbol{\theta} \mid \boldsymbol{Y}] = 0$.
- For $\lambda > 1$, the sign bottleneck obstructs worst-case single-site Glauber mixing.

What was known about posterior sampling?

Work / method	Parameter regime	Guarantee
Montanari–Wu (2023) Diffusion	Sufficiently large fixed λ	Normalized Wasserstein W_2
Paliy–Sohn (2026) Spectral start + Glauber	Sufficiently large fixed λ	Total variation
Bandeira et al. (2026) Glauber	Any fixed $\lambda > 0$, with a *strong extra Gaussian channel*	Total variation for the augmented posterior
This work	Every fixed $\lambda > 1$	Total variation, no supplied side information

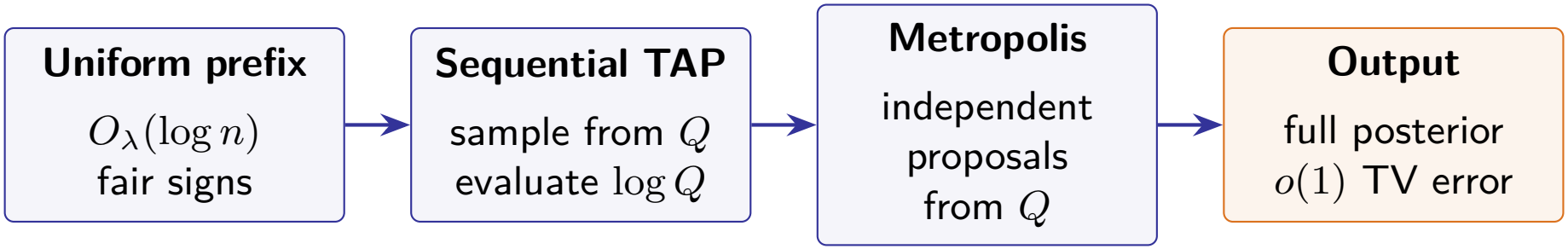
Main result

(L., 2026)

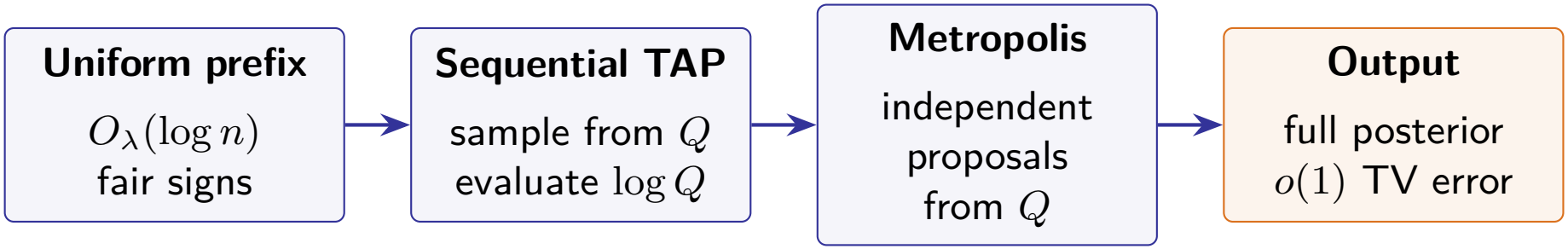
For every fixed $\lambda > 1$, there is a randomized algorithm with running time $n^{O_\lambda(1)}$ whose conditional output law satisfies

$$\mathbb{E}_{\boldsymbol{Y}} \|\widehat{\mu}(\cdot \mid \boldsymbol{Y}) - \mu_{\boldsymbol{Y}}\|_{\mathrm{TV}} \longrightarrow 0.$$

A sequential proposal followed by correction

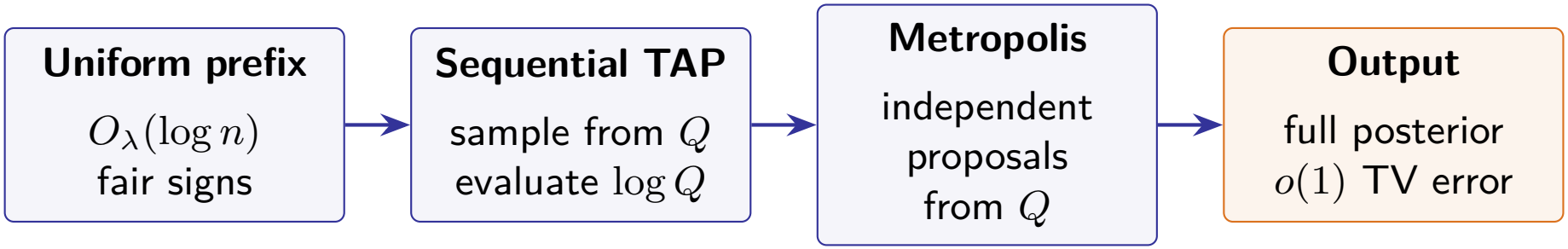


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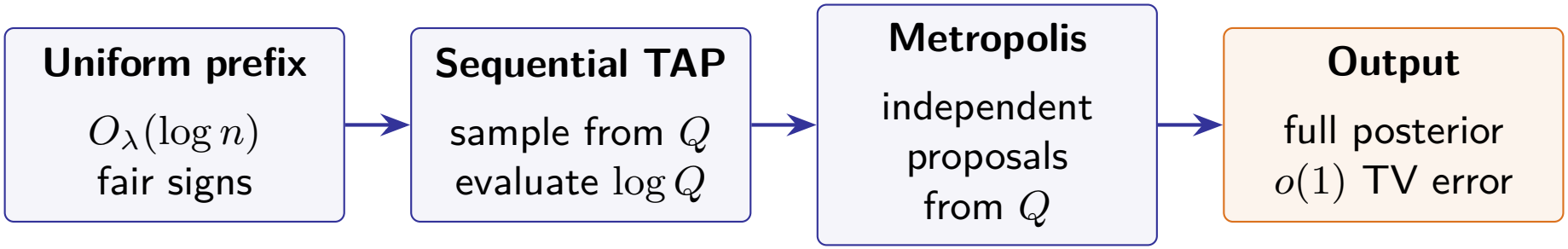
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- Allow a polynomial target-to-proposal density ratio on almost all posterior mass:

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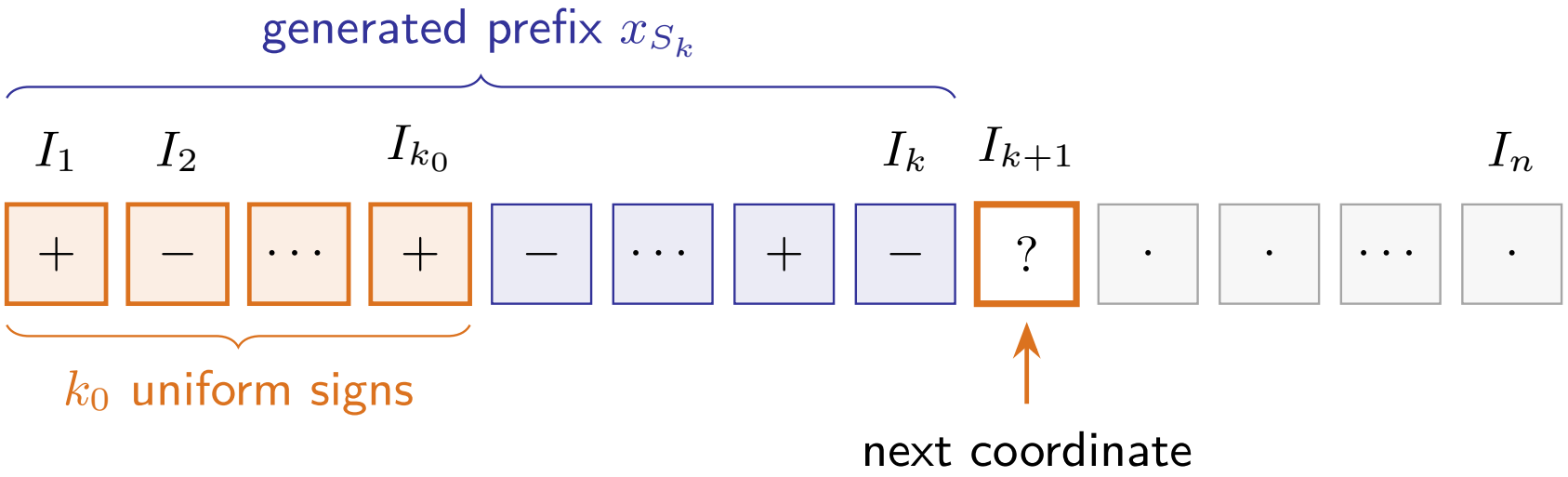


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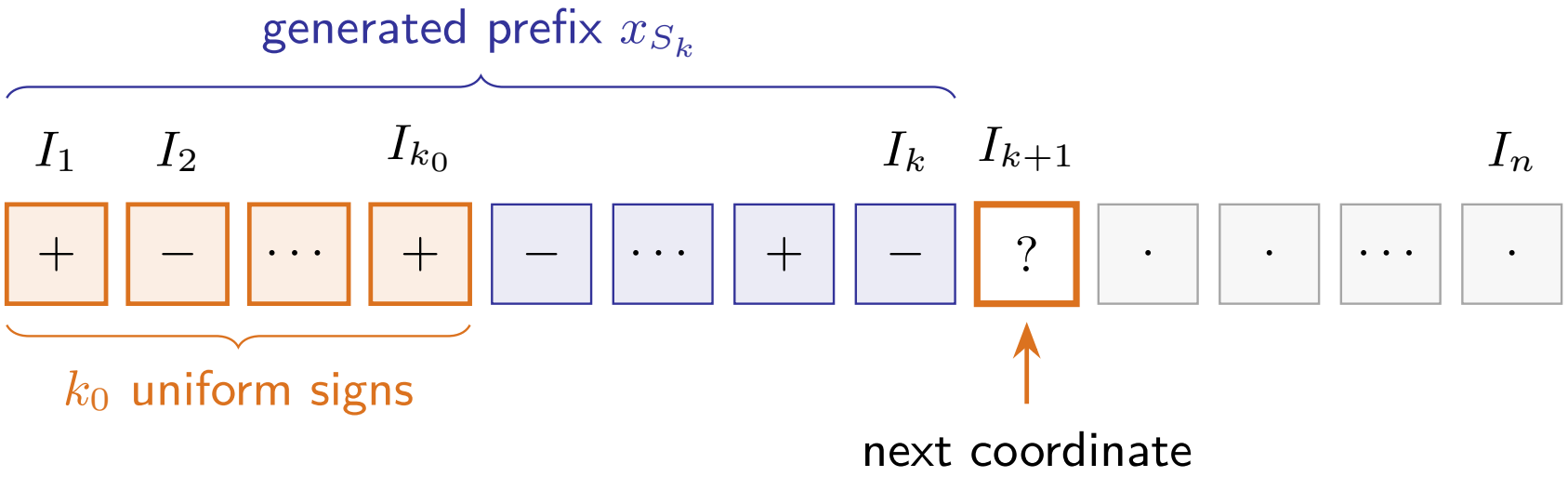
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- Independence Metropolis converts this proposal into a sample with vanishing TV error.

Constructing an evaluable sequential proposal

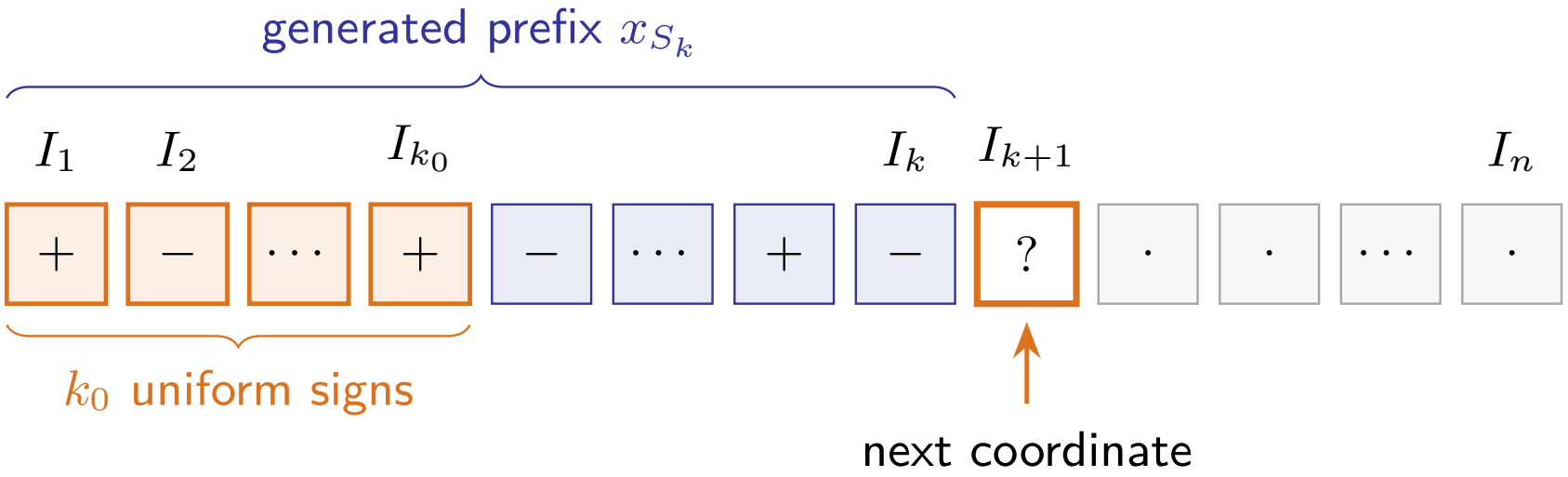


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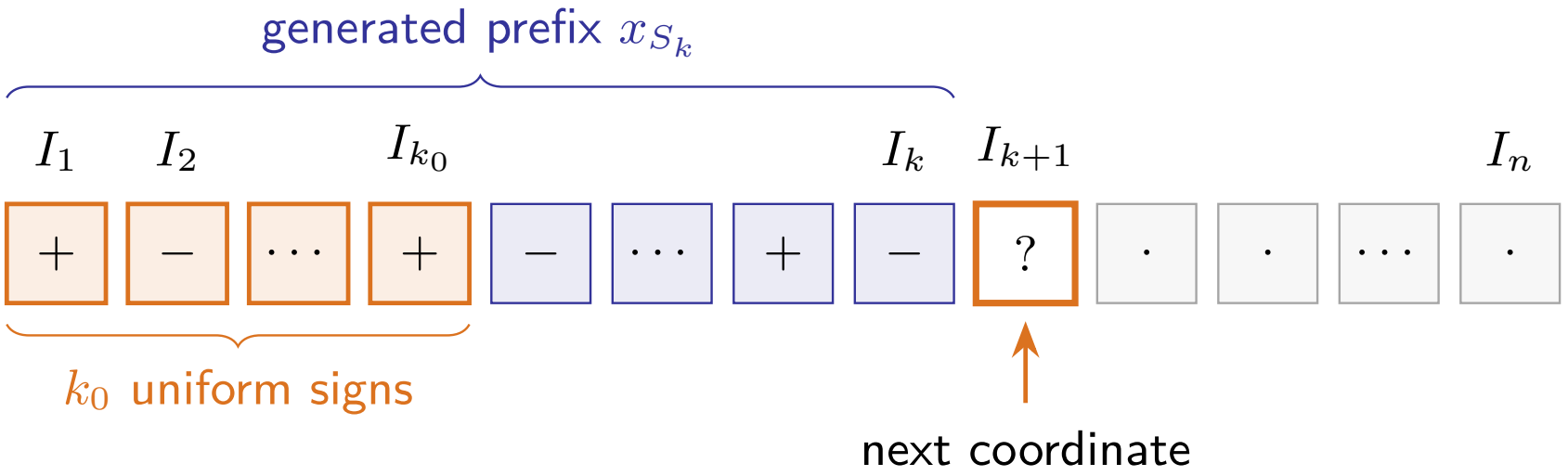
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- Draw the first $k_0 = O_\lambda(\log n)$ signs uniformly.

Constructing an evaluable sequential proposal



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- Draw the first $k_0 = O_\lambda(\log n)$ signs uniformly.
- Given a prefix x_{S_k}, compute the full logit predictor $\widehat{\ell}^{(k)}$, then inspect I_{k+1} and draw its sign.

$$q_k(b \mid x_{S_k}) = \frac{e^{b\widehat{\ell}^{(k)}_{I_{k+1}}}}{2\cosh \widehat{\ell}^{(k)}_{I_{k+1}}}, \qquad Q(x) = 2^{-k_0} \prod_{k=k_0}^{n-1} q_k(x_{I_{k+1}} \mid x_{S_k}).$$

Conditional TAP gives a stable logit predictor

The prefix induces a field $\boldsymbol{h}^{(k)} = \boldsymbol{J}_{R_k,S_k} x_{S_k}$. Thus

$$\mu_{\boldsymbol{J}}(u \mid x_{S_k}) \propto \exp\left\{ \tfrac{1}{2} u^\top \boldsymbol{J}_{R_k,R_k} u + \langle \boldsymbol{h}^{(k)}, u \rangle \right\}.$$

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Write $\boldsymbol{m}^{(k)}$ for its mean and $\ell^{(k)} = \operatorname{arctanh}(\boldsymbol{m}^{(k)})$. Use AMP to choose $\mathcal{K}_k$, then approximately minimize

$$\Phi_k(u) = \sum_{i \in R_k} \phi(u_i) - \tfrac{1}{2} u^\top \boldsymbol{J}_{R_k,R_k} u - \langle \boldsymbol{h}^{(k)}, u \rangle + \tfrac{a_{k/n}}{2} \|u\|^2,$$

Here ϕ is the negative binary entropy.

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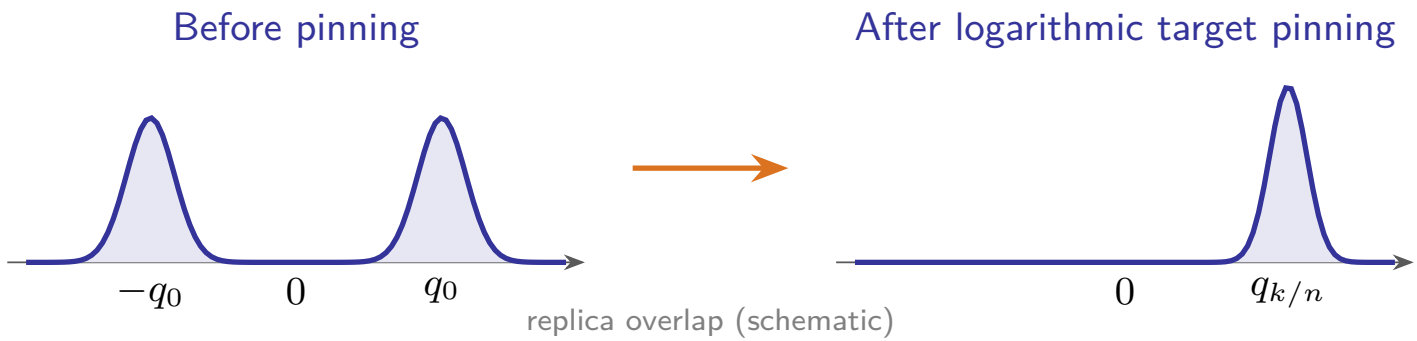
Predict half-log-odds directly

$$\widehat{\ell}^{(k)} = \boldsymbol{h}^{(k)} + (\boldsymbol{J}_{R_k,R_k} - a_{k/n} \boldsymbol{I}) \widehat{\boldsymbol{m}}^{(k)}.$$

This avoids the instability of $\operatorname{arctanh}(\widehat{\boldsymbol{m}}^{(k)})$ near ± 1.

$a_\alpha = \lambda^2(1-\alpha)(1-q_\alpha)$, with the positive fixed point $q_\alpha = \mathbb{E} \tanh^2(s_\alpha + \sqrt{s_\alpha} G)$, $s_\alpha = \lambda^2[(1-\alpha)q_\alpha + \alpha]$, $G \sim \mathcal{N}(0,1)$.

New ingredient 1: logarithmic pinning fixes the phase



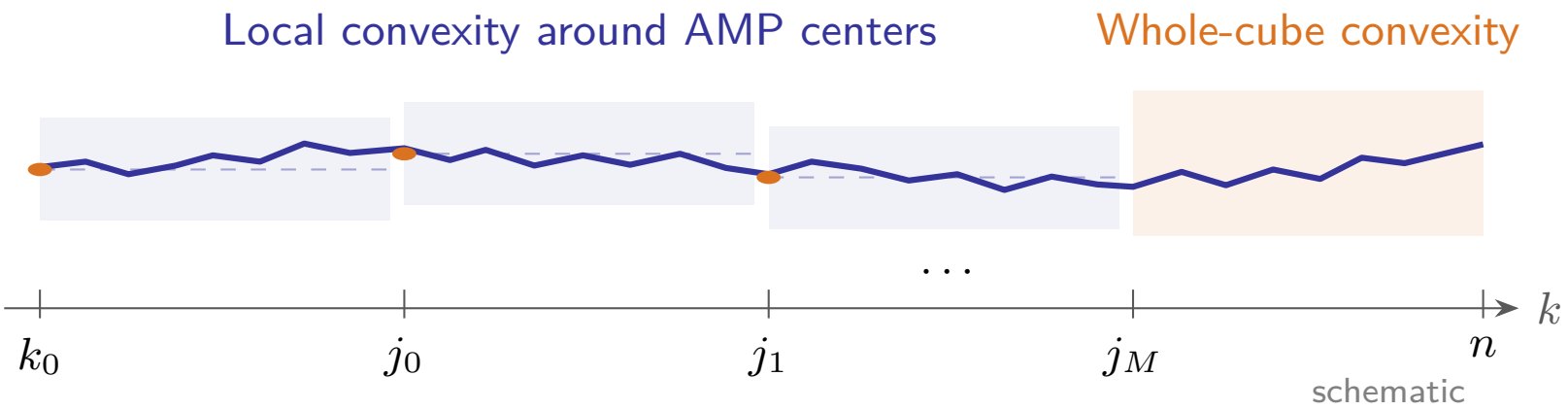
For analysis, reveal $k \geq A_\lambda \log n$ coordinates of a *posterior sample*. Let $N = n - k$ and let σ^1, σ^2 be independent conditional replicas.

Signed-overlap concentration (Proposition 1.2)

$$\mathbb{E}\left\langle \left(\frac{1}{N} \sum_{i \in R_k} \sigma_i^1 \sigma_i^2 - q_{k/n} \right)^2 \right\rangle \leq \frac{C_\lambda}{N}.$$

- Negative-overlap pairs rarely agree on every random pin.
- Absolute-overlap control [EAKJ17] + a new conditioning estimate.

New ingredient 2: convexity along the entire path



- Local TAP convexity at finitely many AMP checkpoints. [CFM23, Cel24]
- The full conditional mean is a martingale:

$$\boldsymbol{M}_k = \mathbb{E}[\boldsymbol{\theta} \mid \boldsymbol{J}, (I_a, \theta_{I_a})_{a \le k}].$$

  A maximal inequality controls motion between checkpoints.
- Transfer Hessian bounds by restriction.

Uniform control along the target path (Lemma 2.4)

With high probability, for all $k \ge k_0$: $\boldsymbol{m}^{(k)} \in \mathcal{K}_k$ and $\nabla^2 \Phi_k(u) \succeq \kappa_\lambda \boldsymbol{I}$ on $\mathcal{K}_k$.

Random order gives an $O(1/n)$ error per step

Pinning + cavity estimates + strong convexity give, on good-prefix events G_k,

$$\mathbb{E}\left[\mathbf{1}_{G_k}\left\|\widehat{\ell}^{(k)}-\ell^{(k)}\right\|^2\right]\le C_\lambda\frac{n-k}{n},\qquad \mathbb{P}\left(\bigcup_{k=k_0}^{n-1}G_k^c\right)=o(1).$$

Both logits are evaluated at the *same posterior-distributed prefix*.

Why compute the vector before inspecting the next index?

Conditionally, I_{k+1} is uniform in R_k, so

$$\mathbb{E}\left[\mathbf{1}_{G_k}|\widehat{\ell}^{(k)}_{I_{k+1}}-\ell^{(k)}_{I_{k+1}}|^2\right]\le\frac{C_\lambda}{n}.$$

For $P_\ell(b)=e^{b\ell}/(2\cosh\ell)$,

$$\mathrm{KL}(P_\ell\|P_{\widehat{\ell}})\le\tfrac{1}{2}(\ell-\widehat{\ell})^2.$$

Summing expected KL errors on good prefixes gives $O_\lambda(1)$.

A bounded entropy budget yields polynomial warmness

- The uniform initial prefix costs at most $2^{k_0} = n^{O_\lambda(1)}$ in the density ratio.
- Stop at the first bad prefix *only in the proof*. The KL chain rule controls the remaining log-likelihood ratio.

Proposal bound (Lemma 2.3)

For $u > 0$, with $b_n = \mathbb{P}(\bigcup_{k \geq k_0} G_k^c) = o(1)$,

$$\mathbb{E}\,\mu_{\boldsymbol{J}}\left(\frac{\mu_{\boldsymbol{J}}(x)}{Q(x)} > 2^{k_0} e^u\right) \leq b_n + \frac{C_\lambda + 1/e}{u}.$$

Take $u = \log n$. Then

$$M = 2^{k_0} n = n^{O_\lambda(1)}, \qquad \mathbb{E}\,\mu_{\boldsymbol{J}}\left(\frac{\mu_{\boldsymbol{J}}}{Q} > M\right) = o(1).$$

Independence Metropolis completes the sampler

Initialize $X_0 \sim Q$. At each step, propose $z \sim Q$ independently and accept with probability

$$\min\left\{1, e^{H_{\boldsymbol{J}}(z)-H_{\boldsymbol{J}}(X_t)}\frac{Q(X_t)}{Q(z)}\right\}.$$

The unknown posterior normalizing constant cancels.

Correction guarantee (Lemma 2.1)

If $\delta = \mu_{\boldsymbol{J}}(\mu_{\boldsymbol{J}}/Q > M)$, then

$$\left\|QK^t - \mu_{\boldsymbol{J}}\right\|_{\mathrm{TV}} \le \delta + \frac{\delta}{M} + \exp\left(-\frac{t(1-\delta)}{M}\right).$$

With $T = \lceil Mn \rceil$ steps, averaging gives $o(1)$ TV error.

Conclusion and future directions

Take-home message

For every fixed $\lambda > 1$, the full $\mathbb{Z}_2$-synchronization posterior admits polynomial-time sampling with vanishing total-variation error.

- New estimates: signed overlap after logarithmic pinning, and conditional TAP control along a random revealing path.
- Algorithmic consequence: an evaluable sequential proposal with polynomial warmness, followed by Metropolis correction.
- Open directions: practical runtimes, uniform control near $\lambda = 1$, and extensions to other planted models.

Thank you!